

Worksheet 9: kNN and k-means

k-nearest neighbor (kNN)

k -nearest neighbor is a simple method to predict the label of new unlabeled observation, given labeled observations. It is a supervised algorithm.

Suppose we have (x_i, y_i) for $i = 1, \dots, n$ where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$ or $y_i \in \{1, 2, \dots, C\}$. That is, the response can be continuous or categorical.

Further, suppose we have an observation x_{new} whose label is not known. Then, we may predict the label of x_{new} by doing choosing in some way between the k -nearest neighbors of x_{new} among x_i . For each x_{new} obtain set of points $S_k(x_{new})$ where

$$S_k(x_{new}) = \{x_i : d(x_{new}, x_i) \text{ is among the } k \text{ smallest distance}\}$$

For continuous response, the predicted y_{new} is

$$y_{new} = \frac{1}{k} \sum_{j: x_j \in S_k} y_j.$$

For a categorical response

$$y_{new} = \arg \max_{c=1,2,\dots,C} \sum_{j: x_j \in S_k} \mathbb{I}(y_j = c).$$

One chooses k by cross-validation typically. Recall in the midesem, many of you essentially did kNN but on the dimension reduced data.

k-means

k -means is an unsupervised clustering algorithm. This means that every observation $x_i \in \mathbb{R}^d$ will look to be clustered into one of k clusters according to a certain criterion. However, the clustering will purely be based on x_i as there will be no label y_i .

Suppose we observe data points

$$x_1, x_2, \dots, x_n \in \mathbb{R}^d.$$

The goal of **clustering** is to partition these observations into groups so that points in the same group are similar to each other and dissimilar to points in other groups. The k -means method is one of the most widely

used clustering procedures. It can be formulated precisely as an **optimization problem**.

Let $k \in \mathbb{N}$ be the desired number of clusters. We introduce two sets of unknown quantities: **cluster centers** (centroids) and **cluster assignments**. Let

$$\mu_1, \mu_2, \dots, \mu_k \in \mathbb{R}^d$$

be the cluster centroids (or means). These are parameters that represent the center of each cluster. Further, for each data point we define an (unobserved) label $z_i \in \{1, 2, \dots, k\}$. The interpretation is

$$z_i = j \quad \text{means point } x_i \text{ belongs to cluster } j.$$

Thus the parameter set representing unknown quantities we want to estimate consists of

$$(\mu_1, \dots, \mu_k, z_1, \dots, z_n).$$

The Fundamental Optimization Problem

The k -means objective measures **within-cluster squared Euclidean distance**. Define

$$J(\mu_1, \dots, \mu_k, z_1, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2.$$

This measures how far each point is from the centroid of the cluster it belongs to. The goal is to find cluster means and cluster assignments such that J is as small as possible. The **k-means optimization problem** is therefore

$$\min_{\mu_1, \dots, \mu_k \in \mathbb{R}^d} \min_{z_1, \dots, z_n \in \{1, \dots, k\}} \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2.$$

Equivalently, we are searching over

- all assignments of points to clusters
- all possible cluster centers

to minimize the total within-cluster squared distance. This objective functions encourages

- points to be close to their assigned centroid
- clusters whose centroids represent their members well.

If we define cluster sets

$$C_j = \{i : z_i = j\},$$

then the objective can also be written

$$J = \sum_{j=1}^k \sum_{i \in C_j} \|x_i - \mu_j\|^2.$$

This highlights that each cluster contributes its **within-cluster variance**.

Non-convexity!

The objective function J is a highly non-convex objective function. This means that there are many local minimas! Thus the optimization itself can be quite challenging, since it is possible that any iterative algorithm only captures a local minima and not a global minima.

You can try to use Nelder-Mead, or other methods, but the best known optimization algorithm is the following.

The k-Means Optimization Algorithm (Lloyd's Algorithm)

Because joint optimization over (μ, z) is combinatorial and difficult, k-means uses an **alternating minimization algorithm**.

1. Choose initial values of the cluster means

$$\mu_1^{(0)}, \dots, \mu_k^{(0)}.$$

2. For each data point compute

$$z_i^{(t)} = \arg \min_j \|x_i - \mu_j^{(t)}\|^2.$$

Each point is assigned to its nearest centroid.

3. For each cluster compute the new centroid

$$\mu_j^{(t+1)} = \frac{1}{|C_j^{(t)}|} \sum_{i \in C_j^{(t)}} x_i.$$

4. Repeat the assignment and update steps until the assignments stop changing or the objective stabilizes.

This algorithm will converge to a minima, but it may not be a global minima, it will be a local minima. Typically, it is a good idea to run the algorithm from different starting values to explore the possibility of improve the minimization. Note this algorithm is only one way to solve the minimization!

Questions

1. Write a function that takes a $n \times d$ matrix X where each row represents a data observation, and implement your own version of the k -means algorithm.
2. Performs k -means on the **faithful** dataset in R, which contains data on the Old Faithful Geyser, with $k = 2$. Provide an interpretation on the resulting clustering.
 - a. First use your own function.

- b. Then use the `kmeans` function.
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- 3. Load the `battingbowling.csv` dataset in R that contains player names, batting averages, bowling averages, and team name.
 - a. Fit $k = 2$ clusters. Based on the clustering, can you provide an interpretation for the clusters?
 - b. Fit $k = 3$ clusters. Provide an interpretation for this clustering.
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- 4. Use k -means to cluster American cities into clusters in the `USArrests` dataset.
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- 5. Load the living room picture provided using `imager`. Considering each pixel as an observation, extract the color palette of this living room using k -means.
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- 6. Load the `digits_truth.Rdata` file which contains the datasets from Question 3 of the exam. Implement kNN on the full dimensions for predictions and compare with dimension reduced prediction error.